Project: Worked examples
Title: Conductor sizing and overload coordination
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

One circuit: the design current, the smallest preferred device rating that carries it, the smallest conductor whose tabulated current-carrying capacity (your own record) carries that rating once the ambient, grouping, installation and semi-enclosed-fuse factors are applied, and the overload coordination checks $I_b \le I_n \le I_z$ and $I_2 \le 1.45\, I_z$. The context (machine wiring or installation) changes the clause citations and the scope note only, never a number.
Design current: $I_b = P/(\sqrt{3}\cdot U\cdot\cos\varphi\cdot\eta)$ three-phase, $P/(U\cdot\cos\varphi\cdot\eta)$ single-phase, P/U DC; or the linked I_b Device: I_n = smallest preferred rating in the ratings record with $I_n \ge I_b$ Fuse factor: C_c = 0.725 for a semi-enclosed fuse (1.45/2, derived), else 1 Required tabulated capacity: $I_{t,req} = I_n/(C_a\cdot C_g\cdot C_i\cdot C_c)$; optional derived $C_a = \sqrt{[(\theta_p - \theta_a)/(\theta_p - \theta_{ref})]}$ (derived approximation — compare with the licensed table) Conductor: CSA = smallest cross-section in the ampacity record with $I_t \ge I_{t,req}$; AWG gauge n: $d = 0.127\cdot 92^{((36 - n)/39)}$ mm, area $= \pi d^2/4$ (derived) Carrying capacity: $I_z = I_t\cdot C_a\cdot C_g\cdot C_i$ Coordination: $I_b \le I_n \le I_z$ and $I_2 \le 1.45\cdot I_z$, $I_2 = 1.45\cdot I_n$ inherent for MCB types B, C, D; for a gG fuse the I_2 condition is deemed satisfied when $I_n \le I_z$; $I_2 = 2\cdot I_n$ by default for a semi-enclosed fuse; otherwise the device's I_2 from its datasheet

Ref: BS 7671:2018 (as amended) reg. 433.1.1 — Co-ordination of conductors and overload protective devices
Ref: BS 7671:2018 (as amended) reg. 523.1 — Current-carrying capacity of a cable
Ref: BS EN 60204-1:2018 cl. 12.4 — Current-carrying capacity of conductors and cables in normal service
Ref: BS EN 60204-1:2018 cl. 7.2 — Overcurrent protection
Ref: BS EN 60898-1 clause number to be checked — Circuit-breakers for overcurrent protection: conventional tripping current
Ref: BS 7671:2018 (as amended) reg. 433.1.201, 433.1.202 — Overload protection by gG fuses and semi-enclosed fuses: deemed conditions

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
context	Design context (changes clause citations and the scope note only)	BS EN 60204-1 machine wiring		spec: Machine design basis MDB-2: BS EN 60204-1 wiring	any
supply	Supply	three-phase		drawing: E-SCH-021 rev A	any
P	Rated output (shaft) power; η converts it to input (DC: electrical power drawn, η not applied)	4	kW	spec: Motor schedule MS-04: conveyor drive 4 kW rated output	0.01 … 500 kW
U	Supply voltage (line voltage for three-phase)	400	V	spec: Site supply: 400 V three-phase	12 … 1000 V
cos_phi	Power factor (not used for DC)	0.82		spec: Motor datasheet MD-04: 0.82 at full load	0.5 … 1
eta	Efficiency (not used for DC)	0.85		spec: Motor datasheet MD-04: 85 % at full load	0.5 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L	Circuit length (carried to the voltage-drop calculation)	35	m	drawing: E-LAY-021 rev B	0.1 … 500 m
theta_a	Ambient temperature	40	degC	spec: Machine design basis MDB-2: 40 °C inside the panel and trunking	-10 … 70 degC
insulation	Insulation	PVC 70 °C		spec: Cable schedule CS-21: PVC 70 °C single cores	any
method	Installation method key (descriptive only, e.g. B2, C, E)	B2		drawing: E-LAY-021 rev B	any
N_g	Grouped circuits	3		drawing: E-LAY-021 rev B	1 … 20
system	Conductor system (AWG: geometry only, no AWG capacities are supplied)	metric		spec: Cable schedule CS-21: metric cross-sections	any
device	Protective device	MCB type C		drawing: E-SCH-021 rev A	any
C_a	Ambient correction factor	0.91		reference: record: owner's licensed table (not shipped); 0.91 is a synthetic value invented for this example	0.1 … 1.5
C_g	Grouping correction factor	0.77		reference: record: owner's licensed table (not shipped); 0.77 is a synthetic value invented for this example	0.1 … 1.5
C_i	Thermal insulation correction factor	1		reference: record: owner's licensed table (not shipped); 1 is a neutral value for this example	0.1 … 1.5
derive_Ca	Derive C_a from the temperatures instead of the entered factor	no		assumption: The entered C_a is used; the derived approximation is not applied.	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
theta_ref	Reference ambient temperature for the derived C_a	30	degC	assumption: Reference ambient of the licensed table; only used when C_a is derived.	0 ... 60 degC
ratings	Preferred device ratings	9 row(s)		reference: record: owner's licensed table (not shipped); ratings are the typical preferred series	any
ratings[1]		I_n = 6 A			
ratings[2]		I_n = 10 A			
ratings[3]		I_n = 16 A			
ratings[4]		I_n = 20 A			
ratings[5]		I_n = 25 A			
ratings[6]		I_n = 32 A			
ratings[7]		I_n = 40 A			
ratings[8]		I_n = 50 A			
ratings[9]		I_n = 63 A			
ampacity	Current-carrying capacities for the installation method	5 row(s)		reference: record: owner's licensed table (not shipped); capacities are synthetic values invented for this example	any
ampacity[1]		CSA = 1.5 mm^2; I_t = 12.3 A			
ampacity[2]		CSA = 2.5 mm^2; I_t = 17.7 A			
ampacity[3]		CSA = 4 mm^2; I_t = 24.1 A			
ampacity[4]		CSA = 6 mm^2; I_t = 31.9 A			
ampacity[5]		CSA = 10 mm^2; I_t = 44.6 A			

3. Assumptions

A1. Every tabulated value (the capacities, the device ratings, the correction factors) is entered from the user's own licensed copy or a manufacturer datasheet; none ships with the calculator. (§A5.)
A2. The capacities in the table are for the installation method stated; the method key is descriptive and is not interpreted. (Records carry their source and edition.)
A3. The ambient, grouping and insulation factors are applied as a product; C_g is not looked up from the number of grouped circuits. (The count is carried to the sheet and warns when C_g = 1 with more than one circuit.)
A4. The optional derived C_a is an approximation for the temperature difference only and is labelled as such; compare it with the licensed table. (Derived, not tabulated.)
A5. C_c = 0.725 applies to a semi-enclosed fuse only: 1.45 divided by an assumed fusing factor of 2. (Derived.)
A6. AWG areas follow the geometry d = 0.127·92^((36 − n)/39) mm; no AWG capacity is supplied. (Derived.)
A7. DC circuits use I_b = P/U; the power factor and efficiency are not applied. (Method.)
A8. derive_Ca (Derive C_a from the temperatures instead of the entered factor) is taken as no. (The entered C_a is used; the derived approximation is not applied.)
A9. theta_ref (Reference ambient temperature for the derived C_a) is taken as 30 degC. (Reference ambient of the licensed table; only used when C_a is derived.)
A10. Machine wiring context (BS EN 60204-1): the conductor is sized for the protective device rating; the capacities and factors are the user's records for the installation method stated. (The context changes citations only; the numbers are the same in both contexts.)

4. Working

```
[14] I_b = P / (sqrt(3) * U * cos_phi * eta)
= 4 kW / (sqrt(3) × 400 V × 0.82 × 0.85)
= 8.2834 A
```
Design current (three-phase)

```
[15] I_n = smallest preferred rating with I_n ≥ I_b
= smallest rating ≥ 8.2834 A in the ratings record
= 10 A
```
BS EN 60204-1 cl. 7.2

Rated current of the protective device (ratings record)

```
[16] C_c = 1 (not a semi-enclosed fuse)
= MCB type C
= 1
```
BS EN 60204-1 cl. 12.4

Semi-enclosed fuse factor

```
[21] I_t_req = I_n / (C_a * C_g * C_i * C_c)
= 10 A / (0.91 × 0.77 × 1 × 1)
= 14.271 A
```
BS EN 60204-1 cl. 12.4

Required tabulated capacity

```
[22] I_t = I_t of the selected row of the ampacity record
= I_t,req = 14.271 A
= 17.7 A
```
BS EN 60204-1 cl. 12.4

Tabulated current-carrying capacity of the selected conductor

```
[23] CSA = smallest cross-section in the ampacity record with I_t ≥ I_t,req
= I_t,req = 14.271 A
= 2.5 mm²
```
BS EN 60204-1 cl. 12.4

Selected conductor cross-section

```
[24] I_z = I_t * C_a * C_g * C_i
= 17.7 A × 0.91 × 0.77 × 1
= 12.402 A
```
BS EN 60204-1 cl. 12.4

Current-carrying capacity in the installed conditions

[25] u_Ib = I_b / I_n — BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1
= 8.2834 A / 10 A
= 0.82834
Design current utilisation of the device

[26] u_In = I_n / I_z — BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1
= 10 A / 12.402 A
= 0.8063
Device rating utilisation of the conductor

[27] I2 = 1.45 * I_n — BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1
= 1.45 × 10 A
= 14.5 A
Conventional operating current (inherent 1.45 I_n)

[28] u_I2 = I2 / (1.45 * I_z) — BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1
= 14.5 A / (1.45 × 12.402 A)
= 0.8063
I_2 utilisation of 1.45 I_z

5. Results

SYMBOL	VALUE	DESCRIPTION
I_b	8.2834 A	Design current
I_n	10 A	Rated current of the device
C_c	1	Semi-enclosed fuse factor
C_a	0.91	Ambient factor
I_t_req	14.271 A	Required tabulated capacity
CSA	2.5 mm²	Selected cross-section
I_z	12.402 A	Current-carrying capacity
L	35 m	Circuit length
u_Ib	0.82834	I_b / I_n
u_In	0.8063	I_n / I_z
I2	14.5 A	Conventional operating current
u_I2	0.8063	I_2 / (1.45 I_z)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Design current within device rating (I_b ≤ I_n)	u_Ib = 0.82834	u_Ib ≤ 1	0.828	BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1
PASS	Device rating within conductor capacity (I_n ≤ I_z)	u_In = 0.8063	u_In ≤ 1	0.806	BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1
PASS	Conventional operating current within 1.45·I_z (I_2 ≤ 1.45·I_z)	u_I2 = 0.8063	u_I2 ≤ 1	0.806	BS EN 60204-1 cl. 7.2; BS 7671 reg. 433.1.1

7. Applicability

No warnings.

8. Verification

Template elec.conductor-sizing@1.0.0 Content hash c40160bd4ebf1da8 Engine calc-engine/1.0.0
Known-answer cases passed 6/6 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0